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Activity Tool : Visualization Evaluation Exercise

[redacted]

I V.P Sneha(192324284) (Name / Reg No) certify that this submission is my original work and that I have adheredto theguidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinaryaction.

Signature of the student: [redacted]

Criteria	**Evaluati of Visualization Quality 25%**	**Critical Analysis 25%**	**Application of Visualization Principles 20%**	**Organization 15%**	**Accuracy 10%**	**Clarity 5%**	**Total**
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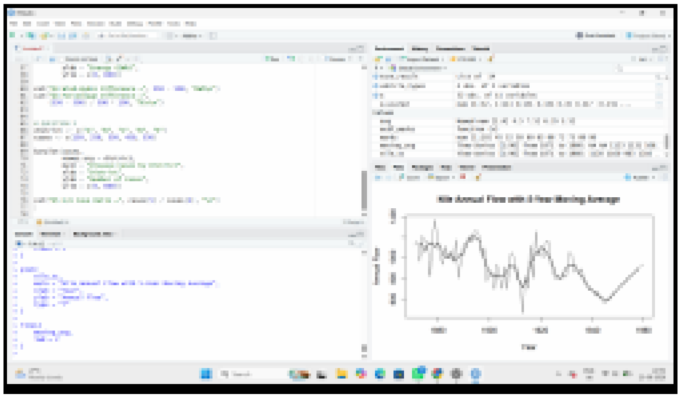

Interpretation

The Nile flow shows noticeable changes in its level over time, with some periods having higher or lower annual flow. The moving average smooths short-term fluctuations and makes longer-term changes in the flowlevel easiertoidentify.

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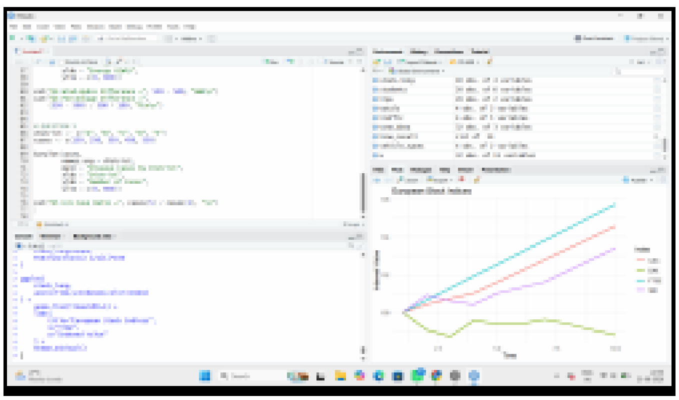

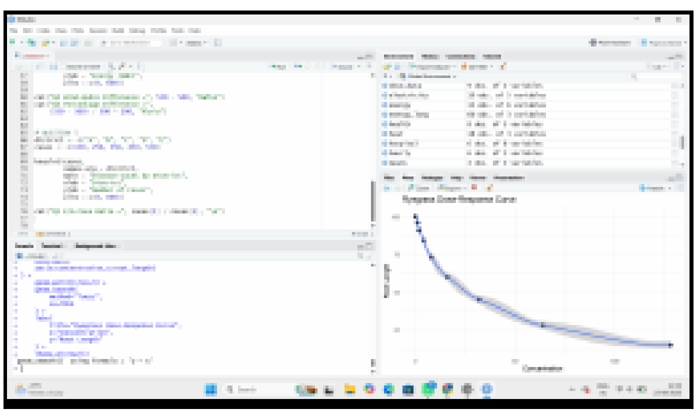

Interpretation

Generally, increasing concentration reduces root length, producing a downward dose-responserelationship. The concentration at which root length is approximately **50% of the control value**can be estimated from the fitted curve.

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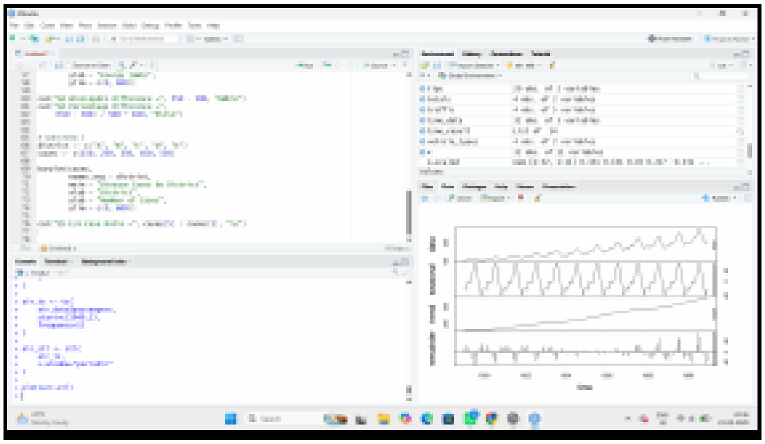
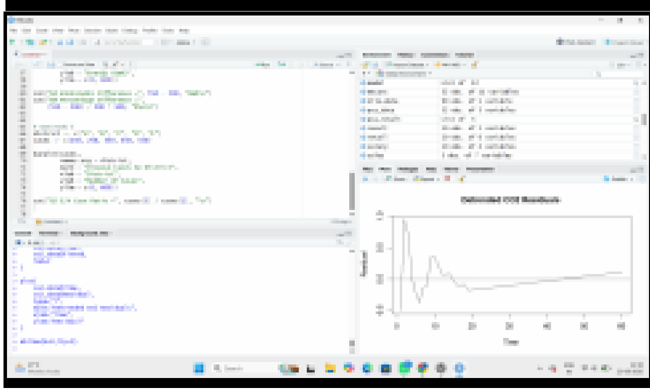


Interpretation

The AirPassengers series has a clear upward trend and strong yearly seasonality. The seasonal fluctuationsbecome larger as the overall number of passengers increases, so a multiplicative decompositionis moreappropriate than a purely additive decomposition.

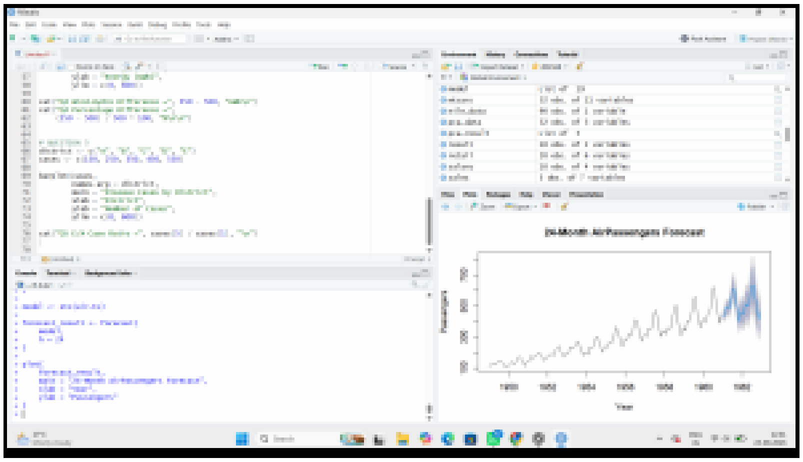
[redacted]



Interpretation

After removing the linear trend, a clear **seasonal/oscillating pattern** remains in the residuals. Therefore, simplelinear detrending is not sufficient because CO2 measurements contain strong seasonal variation in additiontothelong-term upward trend.

[redacted]



Interpretation

The ACF and periodogram show a strong repeating cycle in sunspot activity. The dominant cycleisapproximately 11 years, which agrees closely with the commonly cited solar cycle of about 11years.